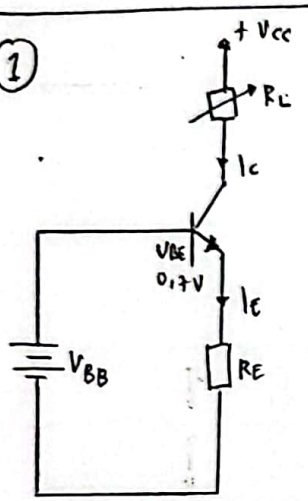
JURNAL ELEKTRONIKA ANALOG LANJUTAN

TRANSISTOR SEBAGAI SUMBER ARUS & SUMBER CERMIN

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Kelas : 2A1 TTE

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① 

Persamaan Garis Beban

$V_{CC} = I_C . R_C + V_{CE} + I_E . R_E \Rightarrow I_C \simeq I_E$

$V_{CC} = I_C . (R_C + R_E) + V_{CE}$

$I_C = \frac{V_{CC} - V_{CE}}{R_C + R_E}$

$I_{csat} = \frac{V_{CC}}{R_C + R_E}$

$I_E = I_C + I_B \rightarrow I_E = I_B (\beta + 1)$
$I_C = \beta . I_B \rightarrow \approx I_B . \beta = I_C .$

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1. Hasil Percobaan untuk VCC = 10 V, RC = 1 kΩ, RE = 470 Ω, VBB = 3 V (RE = 470).

IC	IE	VCE
5,23 mA	5,18 mA	2,46 V

4.

IC	IE	VCE
3,53 mA	5 mA	2,36 V

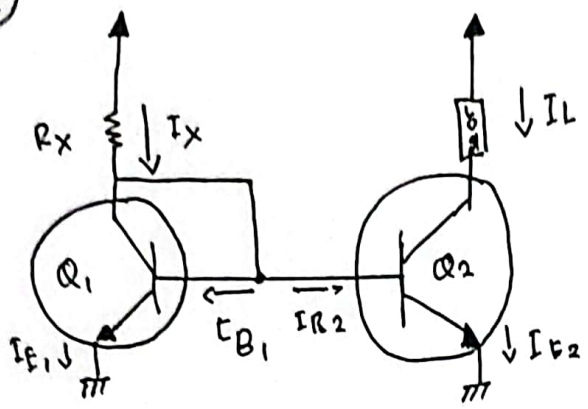
2. Hasil Percobaan untuk VCC = 10 V, RC = 2,2 kΩ, RE = 470 Ω, VBB = 3V (RE = 470)

IC	IE	VCE
5,68 mA	9,09 mA	4,27 V

3.

IC	IE	VCE
2,65 mA	2,65 mA	4,25 V

②



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1. Keterangan: Vcc = 10 V, Rx = 1kΩ, RL = 470Ω, V load = 3 V

Ix	IE2	VCE1	VCE2
8,18 mA	0	0,3 mV	0

2. Keterangan: Vcc = 10V, Rx = 1kΩ, RL = 47Ω, Vload = 3V

Ix	IE2	VCE1	VCE2
8,17mA	0	0,3 mV	0

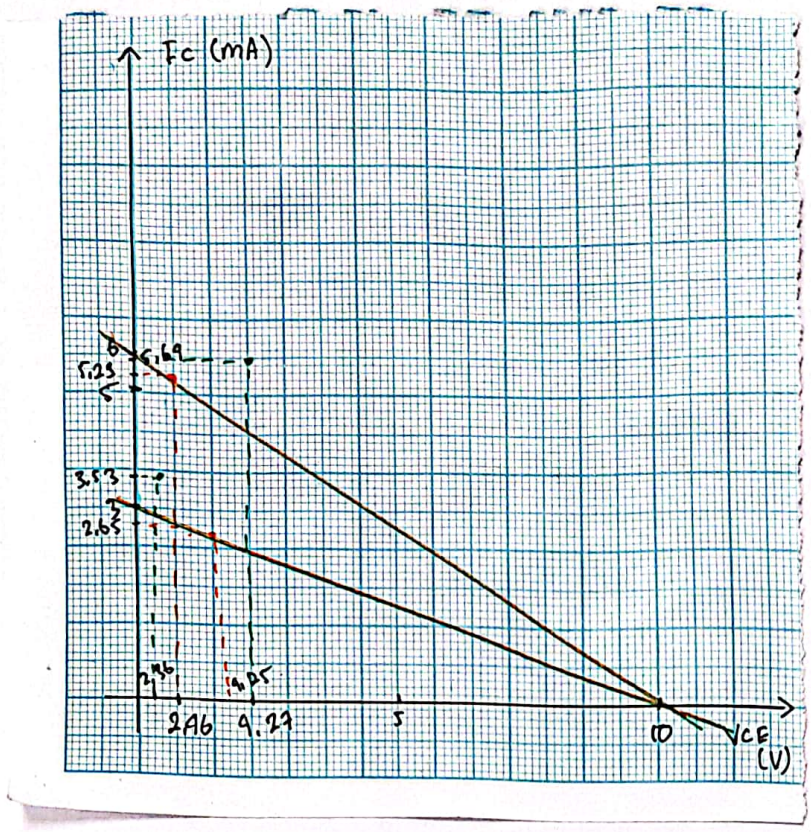
3. Keterangan: Vcc = 10V, Rx = 1KΩ, RL = 470Ω, Vload = V

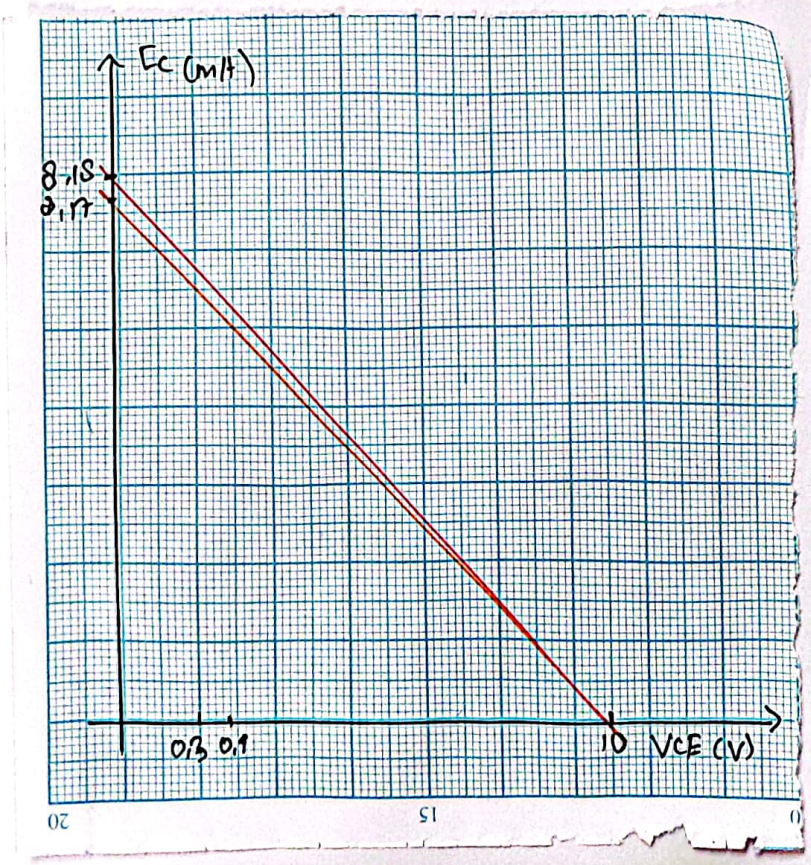
Ix	IE2	VCE1	VCE2
8,18 mA	0	0,3 mV	0

4. Keterangan: Vcc = 10V, Rx = 1kΩ, RL = 470Ω, Vload = 5V

Ix	IE2	VCE1	VCE2
8,17 mA	0	0,4 mV	0

Rangkaian 1


Ic (mA)
6
5,69
5,25
3,53
2,65
2,36
1,25
2A6
9,27
5
10
VCE (V)


Ec (mA)
8,18
2,17
0,3
0,1
10
VCE (V)